Learning objectives

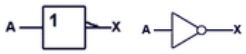
- Logic Circuits
- PLC Operating Principles
- PLC Control Features
- Ladder Diagrams.

Logic Circuits

IEC	ANSI	Description	Boolean
		The AND gate output is at logic 1 when, and only when all its inputs are at logic 1, otherwise the output is at logic 0.	$X = A \cdot B$
		The OR gate output is at logic 1 when one or more of its inputs are at logic 1. If all the inputs are at logic 0, the output is at logic 0.	$X = A+B$
		The NAND Gate output is at logic 0 when, and only when all its inputs are at logic 1, otherwise the output is at logic 1.	$X = \overline{A \cdot B}$
		The NOR gate output is at logic 0 when one or more of its inputs are at logic 1. If all the inputs are at logic 0, the output is at logic 1.	$X = \overline{A+B}$
		The XOR gate output is at logic 1 when one and ONLY ONE of its inputs is at logic 1. Otherwise the output is logic 0.	$X = A \oplus B$
		The XNOR gate output is at logic 0 when one and ONLY ONE of its inputs is at logic 1. Otherwise the output is logic 1. (It is similar to the XOR gate, but its output is inverted).	$X = \overline{A \oplus B}$
		The NOT gate output is at logic 0 when its only input is at logic 1, and at logic 1 when its only input is at logic 0. For this reason it is often called an INVERTER.	$X = \overline{A}$

- International Electrotechnical Commission (IEC)
- American National Standards Institute (ANSI)

Logic Circuits – N/OT (Inverter)



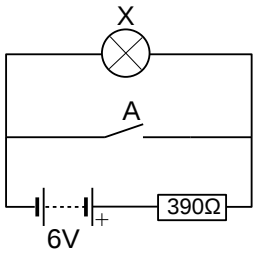
Truth table

A	X | $\overline{A}$
F	T
T	F

Binary Truth table

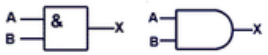
A	X | $\overline{A}$
0	1
1	0

$\overline{A}$ means N/OT A



- Bulb lit when switch is not pressed.
- Bulb is not lit when switch is pressed.
- “X” is the opposite state to “A”.

Logic Circuits - AND



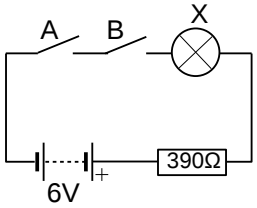
Truth table

A	B	X | A.B
F	F	F
F	T	F
T	F	F
T	T	T

Binary Truth table

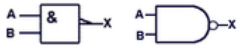
A	B	X | A.B
0	0	0
0	1	0
1	0	0
1	1	1

A.B means “A” AND “B”



- Bulb will only light if “A” AND “B” are depressed together.

Logic Circuits - NAND



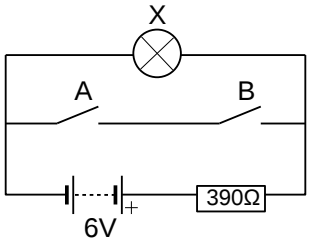
Truth table

A	B	X | $\overline{A.B}$
F	F	T
F	T	T
T	F	T
T	T	F

Binary Truth table

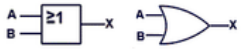
A	B	X | $\overline{A.B}$
0	0	1
0	1	1
1	0	1
1	1	0

$\overline{A.B}$ means N/OT “A” AND “B”



- Bulb will always light unless “A” AND “B” are depressed together.

Logic Circuits - OR



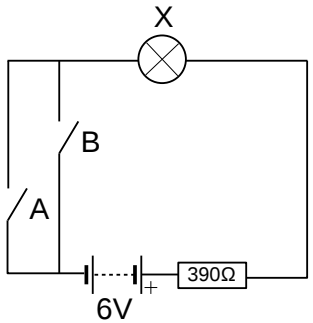
Truth table

A	B	X | A+B
F	F	F
F	T	T
T	F	T
T	T	T

Binary Truth table

A	B	X | A+B
0	0	0
0	1	1
1	0	1
1	1	1

A+B means “A” OR “B”



- Bulb will light if either “A” OR “B” are depressed.

Logic Circuits - NOR

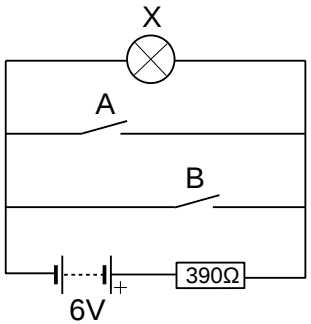
Truth table

A	B	X | $\overline{A+B}$
F	F	T
F	T	F
T	F	F
T	T	F

Binary Truth table

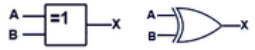
A	B	X | $\overline{A+B}$
0	0	1
0	1	0
1	0	0
1	1	0

$\overline{A+B}$ means “A” N/OR “B”



- Bulb will light unless either “A” or “B” or both are depressed.

Logic Circuits - XOR



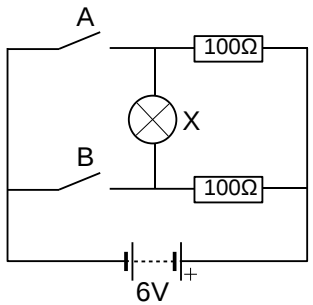
Truth table

A	B	X | A⊕B
F	F	F
F	T	T
T	F	T
T	T	F

Binary Truth table

A	B	X| A⊕B
0	0	0
0	1	1
1	0	1
1	1	0

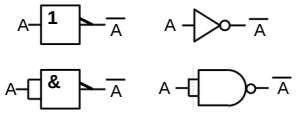
A⊕B means "A" XOR "B"



- Bulb will light if either "A" OR "B" are depressed.

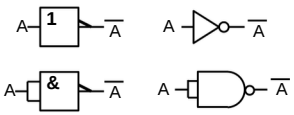
N/OT gates using NAND gates

N/OT



N/OT gates using NAND gates

N/OT



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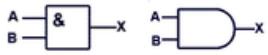
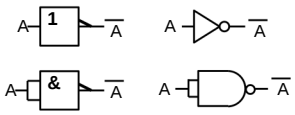


Binary Truth table

A	B	X |A.B
0	0	0
0	1	0
1	0	0
1	1	1

N/OT gates using NAND gates

N/OT



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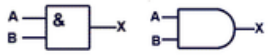
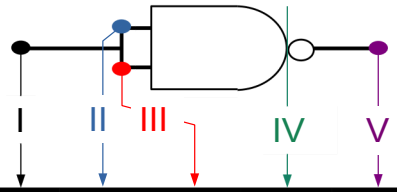


Binary Truth table

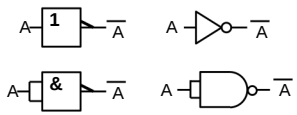
A	B	X |A.B
0	0	0
0	1	0
1	0	0
1	1	1



I	II	III	IV	V
0				
0				
1				
1				

N/OT gates using NAND gates

N/OT



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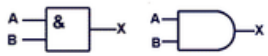
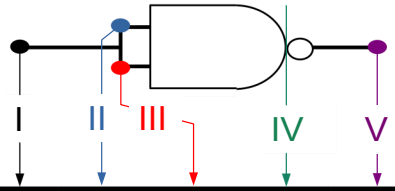


Binary Truth table

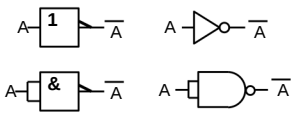
A	B	X |A.B
0	0	0
0	1	0
1	0	0
1	1	1



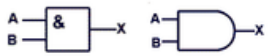
I	II	III	IV	V
0	0			
0	0			
1	1			
1	1			

N/OT gates using NAND gates

N/OT

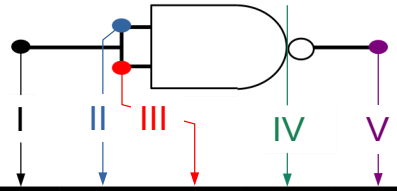


AND



Binary Truth table

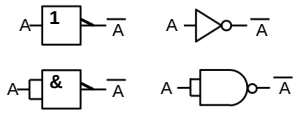
A	B	X |A.B
0	0	0
0	1	0
1	0	0
1	1	1



I	II	III	IV	V
0	0	0		
0	0	0		
1	1	1		
1	1	1		

N/OT gates using NAND gates

N/OT



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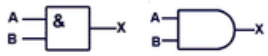
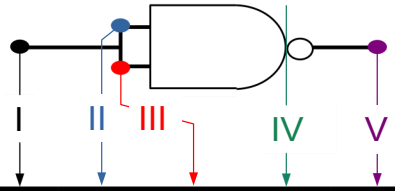


Binary Truth table

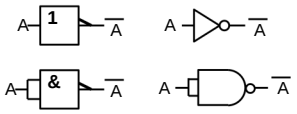
A	B	X |A.B
0	0	0
0	1	0
1	0	0
1	1	1



I	II	III	IV	V
0	0	0	0	
0	0	0	0	
1	1	1	1	
1	1	1	1	

N/OT gates using NAND gates

N/OT



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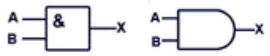
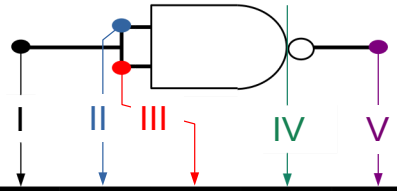


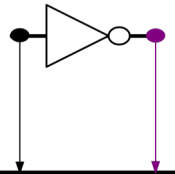
Binary Truth table

A	B	X |A.B
0	0	0
0	1	0
1	0	0
1	1	1

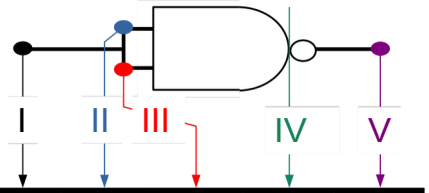


I	II	III	IV	V
0	0	0	0	1
0	0	0	0	1
1	1	1	1	0
1	1	1	1	0

N/OT gates using NAND gates



A	X | $\overline{A}$
0	1
1	0



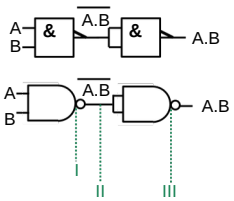
I	II	III	IV	V
0	0	0	0	1
0	0	0	0	1
1	1	1	1	0
1	1	1	1	0

AND gates using NAND gates

AND

A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

AND from NAND gates



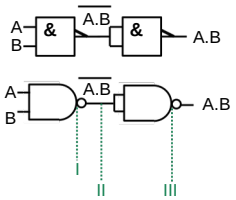
A	B	I	II	III	A.B
0	0				
0	1				
1	0				
1	1				

AND gates using NAND gates

AND

A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

AND from NAND gates



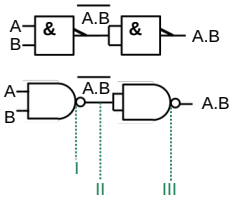
A	B	I	II	III	A.B
0	0	0			
0	1	0			
1	0	0			
1	1	1			

AND gates using NAND gates

AND

A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

AND from NAND gates



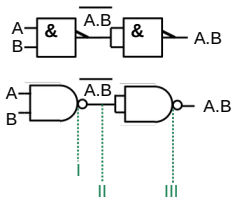
A	B	I	II	III	A.B
0	0	0	1		
0	1	0	1		
1	0	0	1		
1	1	1	0		

AND gates using NAND gates

AND

A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

AND from NAND gates



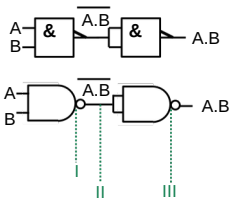
A	B	I	II	III	A.B
0	0	0	1	1	
0	1	0	1	1	
1	0	0	1	1	
1	1	1	0	0	

AND gates using NAND gates

AND

A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

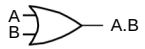
AND from NAND gates



A	B	I	II	III	A.B
0	0	0	1	1	0
0	1	0	1	1	0
1	0	0	1	1	0
1	1	1	0	0	1

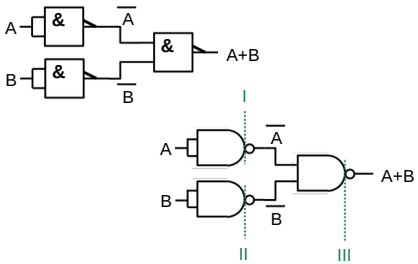
OR gates using NAND gates

OR



A	B	A.B
0	0	0
0	1	1
1	0	1
1	1	1

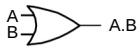
OR from NAND gates



A	I	$\overline{A}$	B	II	$\overline{B}$	III	A+B
0			0				
0			1				
1			0				
1			1				

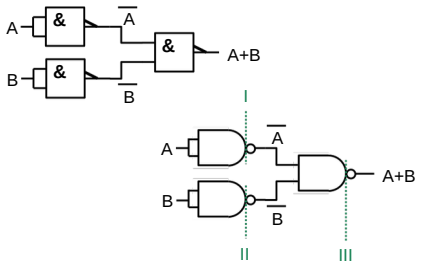
OR gates using NAND gates

OR



A	B	A.B
0	0	0
0	1	1
1	0	1
1	1	1

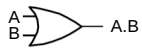
OR from NAND gates



A	I	$\overline{A}$	B	II	$\overline{B}$	III	A+B
0	0	1	0				
0	0	1	1				
1	1	0	0				
1	1	0	1				

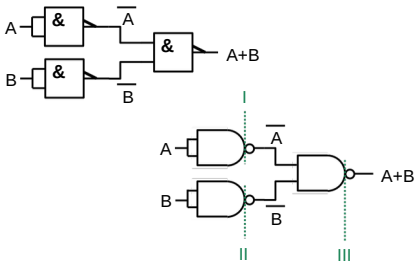
OR gates using NAND gates

OR



A	B	A.B
0	0	0
0	1	1
1	0	1
1	1	1

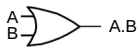
OR from NAND gates



A	I	$\overline{A}$	B	II	$\overline{B}$	III	A+B
0	0	1	0	0	1		
0	0	1	1	1	0		
1	1	0	0	0	1		
1	1	0	1	1	0		

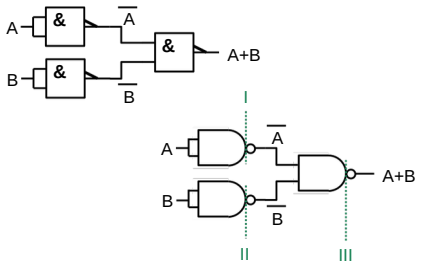
OR gates using NAND gates

OR



A	B	A.B
0	0	0
0	1	1
1	0	1
1	1	1

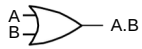
OR from NAND gates



A	I	$\overline{A}$	B	II	$\overline{B}$	III	A+B
0	0	1	0	0	1	1	
0	0	1	1	1	0	0	
1	1	0	0	0	1	0	
1	1	0	1	1	0	0	

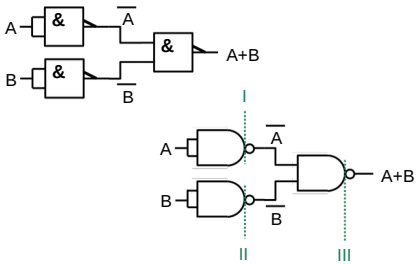
OR gates using NAND gates

OR



A	B	A.B
0	0	0
0	1	1
1	0	1
1	1	1

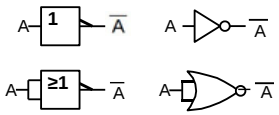
OR from NAND gates



A	I	$\overline{A}$	B	II	$\overline{B}$	III	A+B
0	0	1	0	0	1	1	0
0	0	1	1	1	0	0	1
1	1	0	0	0	1	0	1
1	1	0	1	1	0	0	1

All gates using NOR gates

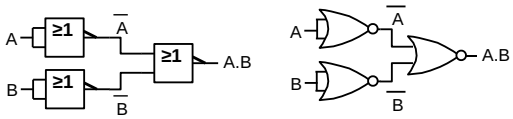
N/OT



Binary Truth table

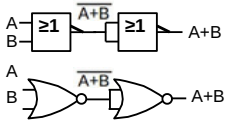
A	B	$\overline{A+B}$
0	0	0
0	1	1
1	0	1
1	1	1

AND



A	B	$\overline{A}$	$\overline{B}$	A.B
0	0	1	1	0
0	1	1	0	0
1	0	0	1	0
1	1	0	0	1

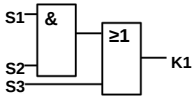
OR



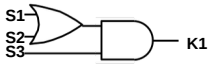
A	B	$\overline{A+B}$	A+B
0	0	1	0
0	1	0	1
1	0	0	1
1	1	0	1

Exercise #1

1. Draw the equivalent circuit for the following logic gates

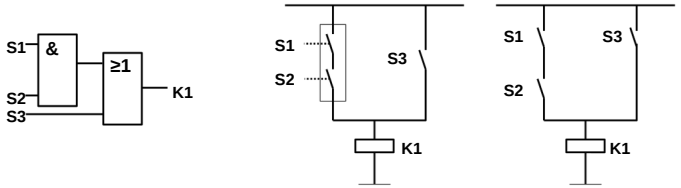


2. Draw the equivalent circuit for the following logic gates

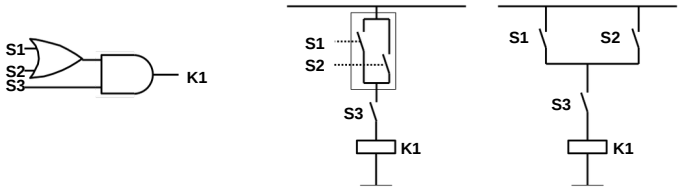


Exercise #1

1. Draw the equivalent circuit for the following logic gates

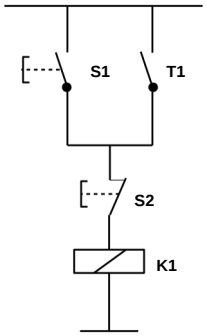


2. Draw the equivalent circuit for the following logic gates



Exercise #2

Draw the logic gate combination to represent the following circuit diagram



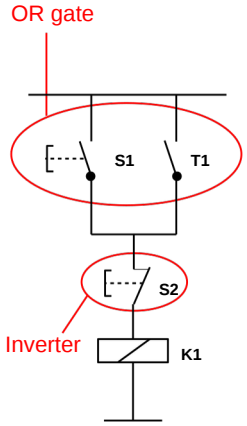
Truth table

2^3	S1	T1	S2			K1
0						0
1						0
2						1
3						0
4						1
5						0
6						1
7						0

S2 button not pressed conducts, button pressed short-circuit so it acts like an inverter.

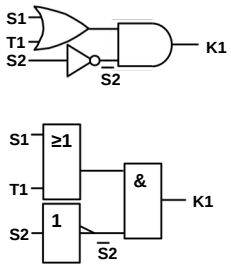
Exercise #2

Draw the logic gate combination to represent the following circuit diagram



Truth table

2^3	S1	T1	S2	X|OR	$\overline{S2}$	K1
0	0	0	0	0	1	0
1	0	0	1	0	0	0
2	0	1	0	1	1	1
3	0	1	1	1	0	0
4	1	0	0	1	1	1
5	1	0	1	1	0	0
6	1	1	0	1	1	1
7	1	1	1	1	0	0



S2 button notpressed conducts, button pressed short-circuit so it acts like an inverter.